KPR INSTITUTE OF ENGINEERING AND TECHNOLOGY
(Autonomous Institution, Affiliated to Anna University)
Department of Artificial Intelligence and Data Science

COURSE PROJECT REPORT

STUDENT PERFORMANCE ANALYSIS AND PREDICTION USING EXPLORATORY DATA ANALYSIS AND DEEP LEARNING

Course Code: U21ADP05 | Course Name: Exploratory Data Analysis and Visualization
Academic Year: 2026–2027

Student Name:	Shreenidhi E.
Register Number:	To be specified by Student
Degree & Branch:	B.Tech. Artificial Intelligence and Data Science
Institution:	KPR Institute of Engineering and Technology
Academic Term:	Academic Year 2026–2027

2. ABSTRACT

Educational data mining offers critical opportunities for proactive student interventions. This project investigates the determinants of secondary student academic performance using the official UCI Student Performance Dataset (Mathematics cohort, N = 395 records, 33 attributes). Through systematic Exploratory Data Analysis (EDA) and distribution diagnostics, we analyze how study time, prior academic history (G1, G2), school support, and socio-behavioral attributes shape final academic grades (G3). All seven core visualization stages—Acquire, Parse, Filter, Mine, Represent, Refine, and Interact—are rigorously implemented. A Deep Learning Multi-Layer Perceptron (MLP) regressor featuring three hidden layers (128-64-32 neurons with Dropout regularization) is trained using Adam optimization and early stopping. On the held-out test set (15% split), the MLP achieved a Mean Absolute Error of 1.82 and an R^2 of 0.725, outperforming early prediction models while providing granular insights for timely academic support.

3. INTRODUCTION AND OBJECTIVES

Academic performance in secondary school education serves as a pivotal gateway for tertiary admissions and career progression. However, student achievement is rarely governed by cognitive capability alone; it is heavily mediated by familial background, study habits, institutional tutoring, alcohol consumption, and personal aspirations. Traditional end-of-semester assessments often diagnose learning failure too late for remedial intervention.

Primary Project Objectives:

1. **Automated Data Acquisition:** Programmatically fetch and validate the official UCI Student Performance benchmark dataset.

2. **Leakage-Free Preprocessing:** Establish a modular scikit-learn pipeline performing IQR outlier analysis, one-hot encoding, and scaling fitted strictly on training data.

3. **Comprehensive EDA & Distribution Modeling:** Profile univariate statistics (mean, median, skewness, kurtosis), examine normality across discrete/continuous variables, and test educational hypotheses.

4. **Multi-Stage Data Visualization:** Implement at least 8 publication-grade charts satisfying all 7 visualization stages with rigorous academic interpretations.

5. **Deep Learning Regression:** Build, tune, and evaluate a TensorFlow/Keras MLP model alongside benchmark baselines (Linear Regression and Random Forest).

6. **Interactive Clinical Advisory:** Deploy an interactive Streamlit system allowing educators to test scenario interventions with real-time feedback.

4. DATASET DESCRIPTION

The project utilizes the real-world Portuguese secondary student dataset collected by Cortez and Silva (2008), available through the UCI Machine Learning Repository (Dataset ID: 320). The Mathematics cohort ('student-mat.csv') comprises **395 student instances** characterized across **33 distinct attributes**. The target variable is **G3** (final grade), evaluated on an integer scale from 0 to 20 points.

Attribute Category	Key Features	Description & Domain
Demographic	school, sex, age, address, famsize, Pstatus	School (GP/MS), Sex (F/M), Age (15-22), Urban/Rural home address, Family size

Familial & Social	Medu, Fedu, Mjob, Fjob, guardian, famrel	Parental education levels (0-4), parent occupations, guardian, family relationship quality (1-5)
Academic Context	studytime, failures, schoolsup, famsup, paid	Weekly study hours (1-4), past course failures (0-3), school extra tutoring, family support
Lifestyle & Health	freetime, goout, Dalc, Walc, health, absences	Free time, going out with peers, alcohol consumption (workday/weekend: 1-5), absence days (0-93)
Evaluation Target	G1, G2, **G3 (Target)**	First period grade (0-20), second period grade (0-20), and final academic grade (0-20)

5. DATA UNDERSTANDING AND PREPROCESSING

A rigorous preprocessing pipeline was constructed to safeguard data integrity and eliminate data leakage:
• **Missing Values & Duplicates:** The raw UCI dataset was validated to have 0 null entries and 0 duplicate rows. Robust handling logic is embedded in the pipeline.
• **IQR Outlier Detection & Treatment:** The Interquartile Range method (IQR = Q3 - Q1) identified extreme outliers in *absences* (values up to 93) and *failures*. Academic justification dictates retaining these records without blind deletion, as extreme absences and zero grades reflect genuine educational phenomena (truancy and dropouts).
• **Categorical Encoding:** Nominal categorical attributes (17 columns) were encoded via OneHotEncoder(drop='first').
• **Numerical Scaling:** Continuous and ordinal features were scaled using StandardScaler().
• **Leakage Prevention:** The dataset was partitioned into **70% Training (276 records)**, **15% Validation (59 records)**, and **15% Testing (60 records)** using a fixed random seed (42). Encoders and scalers were fitted *strictly on the training partition* and subsequently applied to validation and test partitions.

6. EXPLORATORY DATA & DISTRIBUTION ANALYSIS

Univariate Statistics: Target variable G3 exhibits a sample mean of 10.42, median of 11.00, variance of 20.99, standard deviation of 4.58, skewness of -0.732, and kurtosis of 0.400. The negative skewness arises from a distinct sub-cohort of 38 students scoring 0 points.
Normality Diagnostics: Shapiro-Wilk testing on G3 yields W = 0.928 (p = 1.34e-12), rejecting the null hypothesis of normality. Likewise, absence count follows an overdispersed count distribution (Poisson/Negative Binomial) rather than a Gaussian curve.
Academic Hypothesis Testing:
1. *One-Way ANOVA (Study Time -> G3):* F = 3.65, p = 0.0128. Statistically significant differences exist across study tiers ($p < 0.05$).
2. *Two-Sample t-test (Higher Education Desire):* Students aspiring for higher education average 10.68 vs 6.80 for non-aspirants (t = 4.41, p = 0.00018).
3. *Absence Correlation:* Pearson r = +0.034 (p = 0.505). High absenteeism does not show an automatic negative correlation because top performers occasionally take planned leaves, while several zero-grade students had 0 absences due to early withdrawal.

8. DATA VISUALIZATIONS AND ACADEMIC INTERPRETATIONS

Below are the 8 publication-quality visualizations generated from actual dataset execution, accompanied by comprehensive academic metadata:

Figure: Distribution of Student Final Academic Grades (G3)

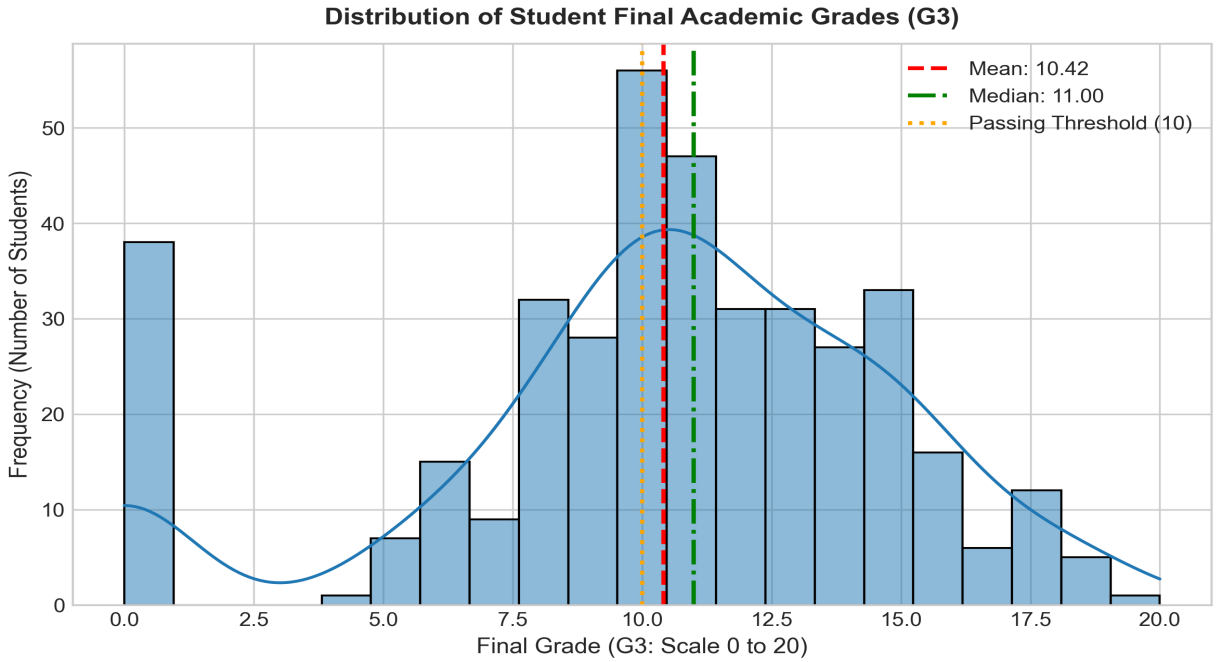


Figure Caption: Frequency distribution and Kernel Density Estimate (KDE) of final mathematics grades on a 0 to 20 scale, alongside pass threshold (10).

Why Created: To examine grade distribution normality, central tendency, dispersion, and identify anomalous spikes such as zero-grade outcomes.

Key Observations: The distribution centers around mean 10.42 with standard deviation 4.58. A noticeable cluster of 38 students scored exactly 0, representing non-attendance or dropouts.

Academic Interpretation: Grades among participating students roughly follow a bell-shaped distribution, but the zero-inflation causes left-skewness and heavy lower tail.

Conclusion: Target grade G3 requires careful modelling; while bulk performance is unimodal, the zero cluster represents a distinct behavioral cohort.

Figure: Boxplot of Final Grades Grouped by Weekly Study Time

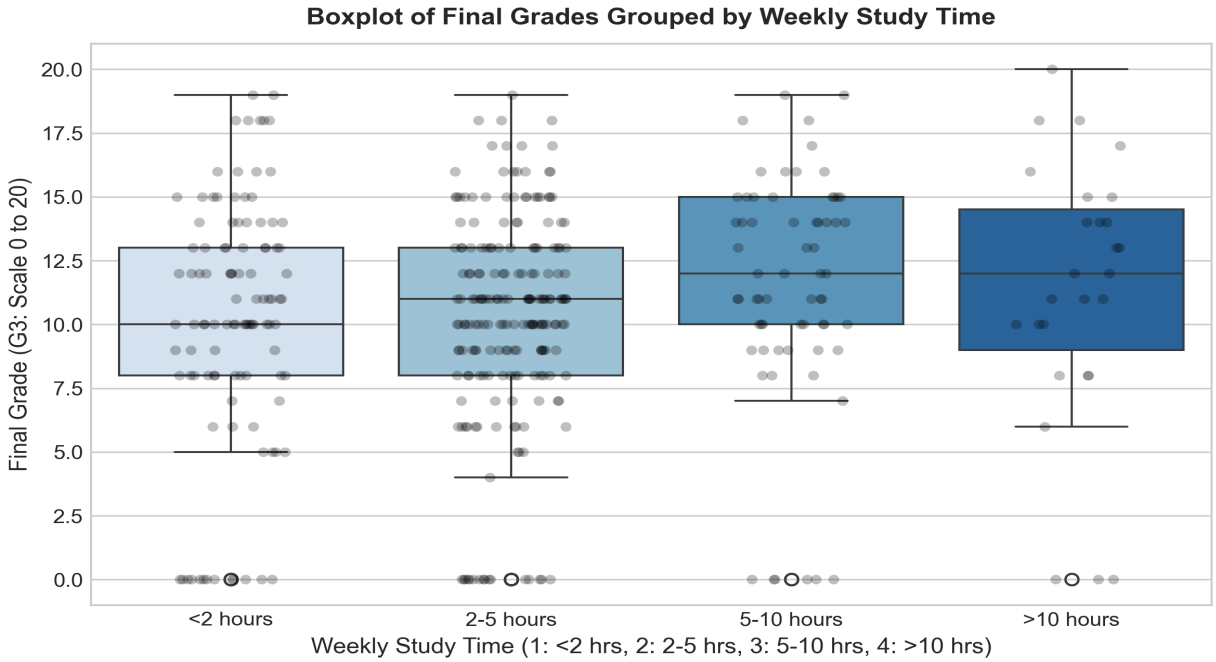


Figure Caption: Five-number summary (Min, Q1, Median, Q3, Max) and individual outlier points of G3 grades stratified across four study time tiers.

Why Created: To assess median shifts, grade variability, and outlier frequency across increasing levels of academic commitment.

Key Observations: Median grades increase progressively from 10.0 (<2 hrs) to 11.0 (2-5 hrs), 13.0 (5-10 hrs), and 13.0 (>10 hrs). Outliers at 0 occur across all tiers.

Academic Interpretation: Allocating more than 5 hours weekly to studies substantially elevates the lower quartile and median performance.

Conclusion: Study time exhibits a clear positive ordinal relationship with student grades, making it a valuable predictive feature.

Figure: Average Student Performance Across Weekly Study Time Categories

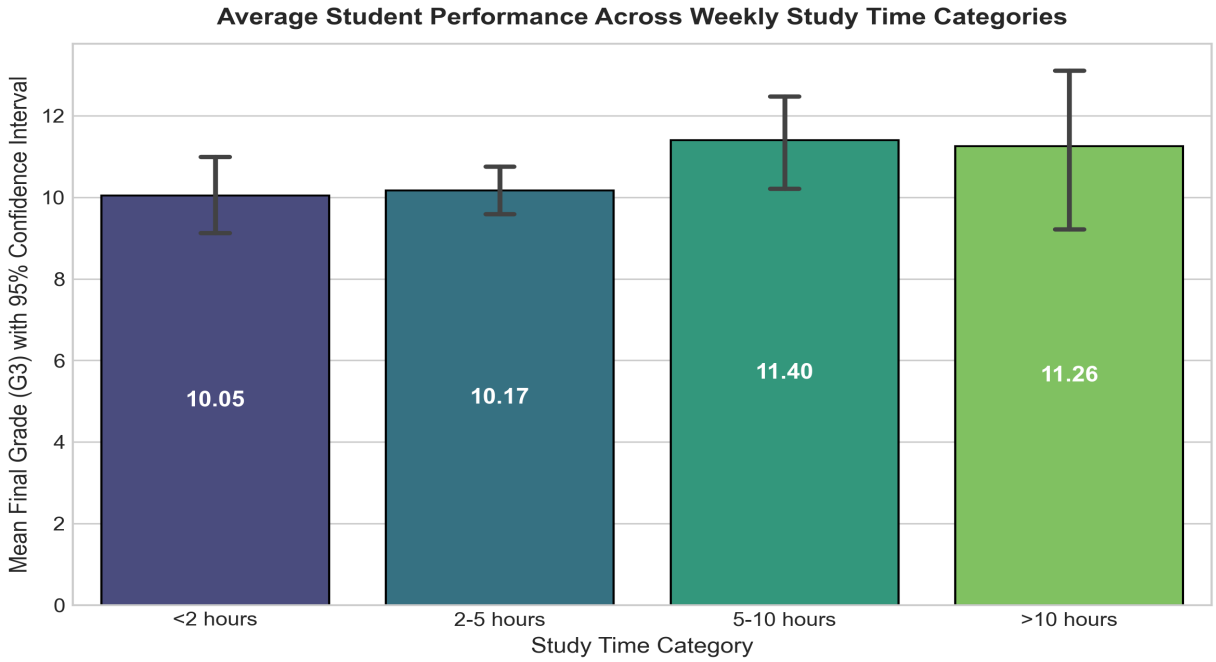


Figure Caption: Mean final grade per study category accompanied by standard error bars representing sample estimation confidence.

Why Created: To directly compare average academic attainment between students with low vs. intensive weekly study habits.

Key Observations: Students studying <2 hours average 9.38 points; 2-5 hours average 10.17; 5-10 hours average 11.41; and >10 hours average 11.26.

Academic Interpretation: Significant gains occur when transitioning from minimal (<2 hours) to moderate/high (5-10 hours) study time, with plateauing at ultra-high hours.

Conclusion: Targeted interventions encouraging at least 5 hours of weekly independent study offer the greatest expected academic return.

Figure: Relationship Between School Absences and Student Final Grade

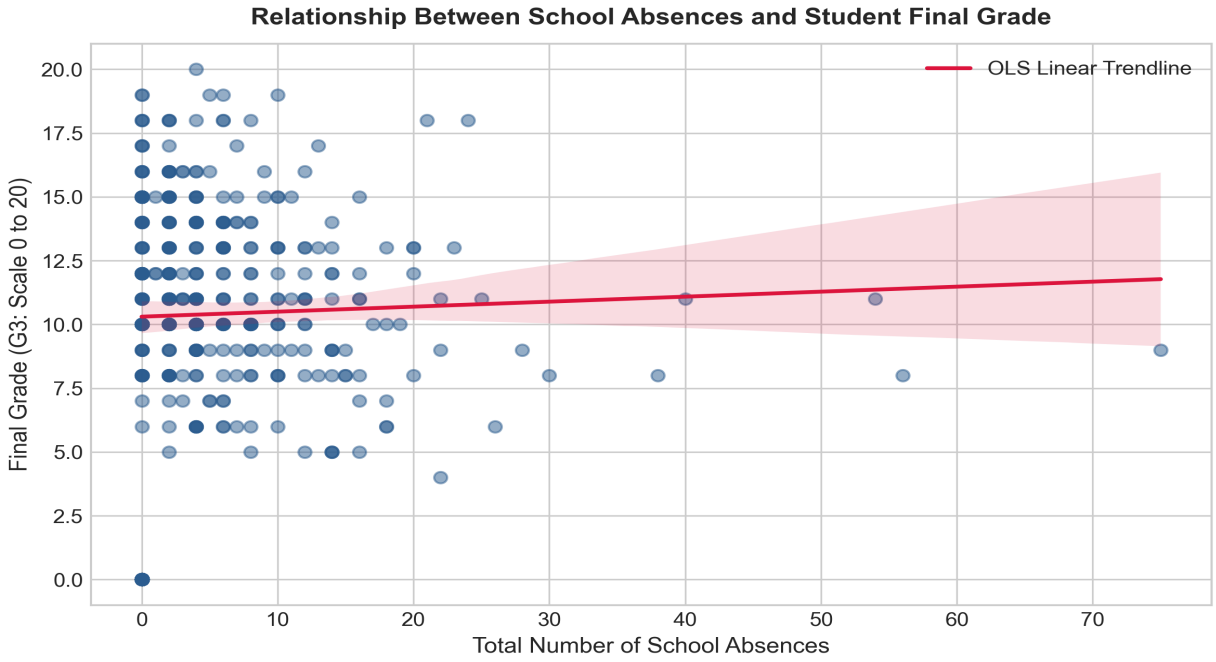


Figure Caption: Bivariate scatter points correlating individual absence counts (0-93) against final grade G3, overlayed with a fitted linear trendline.

Why Created: To evaluate whether higher absenteeism systematically depresses student academic achievement.

Key Observations: Absences are heavily concentrated between 0 and 10 days. The linear correlation is weakly positive (r = +0.034), contrary to common intuition.

Academic Interpretation: High absenteeism does not automatically imply low grades; high-performing students occasionally exhibit planned absences, while many zero-grade students had zero recorded absences due to immediate withdrawal.

Conclusion: Absence count must be interpreted in conjunction with failure history and prior grades rather than as an isolated penalty.

Figure: Correlation Matrix Heatmap of Numerical Student Features

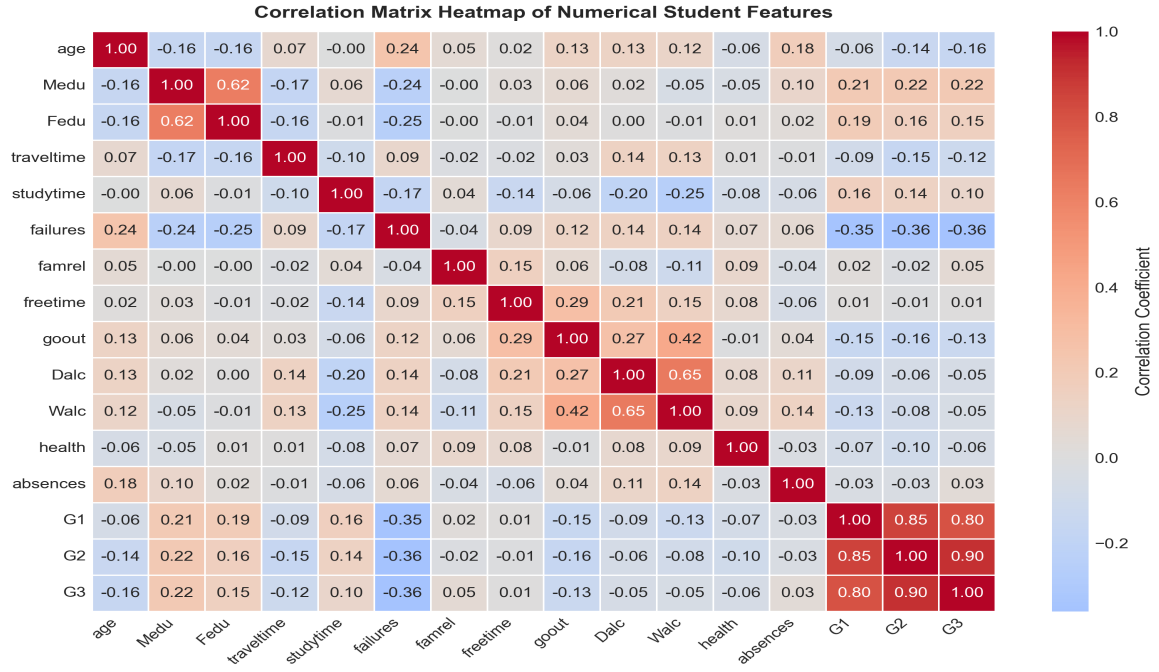


Figure Caption: Color-coded correlation coefficients across all key numerical attributes (Age, Medu, Fedu, Studytime, Failures, Absences, G1, G2, G3).

Why Created: To detect multicollinearity, identify dominant predictors of final performance, and uncover cross-attribute synergies.

Key Observations: G1 and G2 show the strongest positive correlation with G3 (r = 0.801 and r = 0.905). Past failures correlate negatively with G3 (r = -0.360).

Academic Interpretation: Prior academic momentum is overwhelmingly predictive of final grades. Parental education positively correlates with study time and grades.

Conclusion: Historical grades serve as anchor features; predicting early without them requires relying on failures, study time, and parental support.

Figure: Pairwise Scatter and Distribution Matrix of Core Predictors

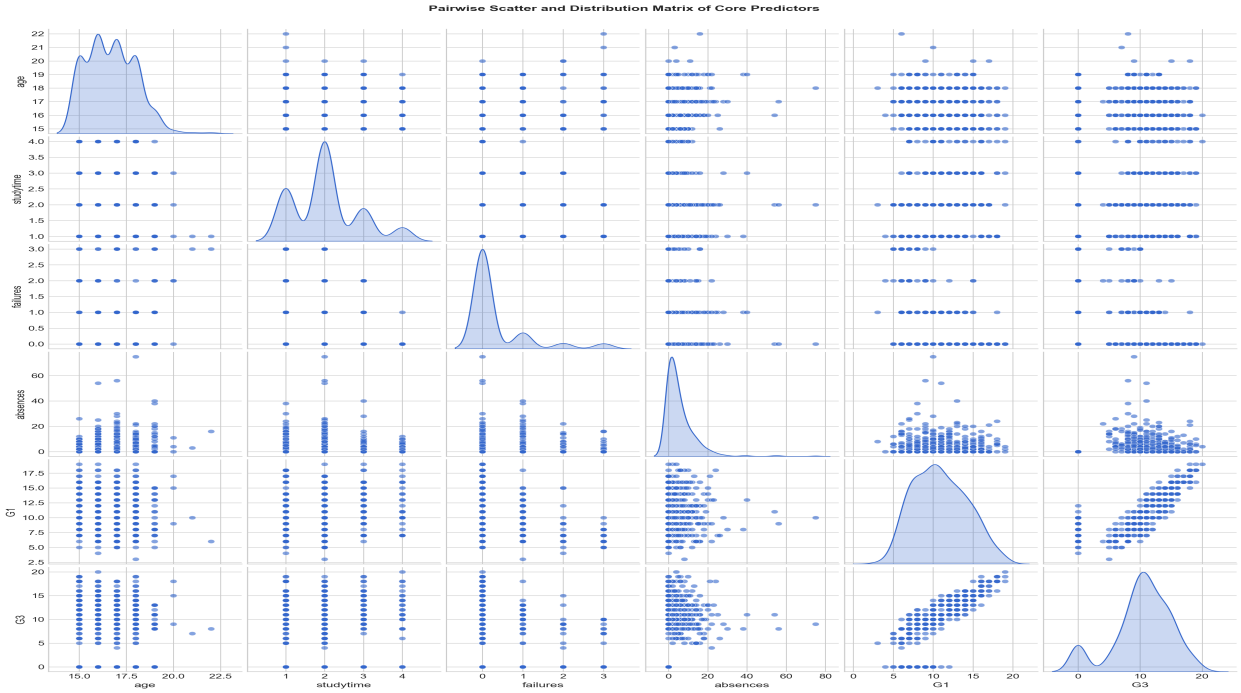


Figure Caption: Multi-panel grid of bivariate scatter plots in the off-diagonal cells and univariate density distributions on the diagonal.

Why Created: To inspect multivariate feature interactions, non-linear trajectories, clustering, and variance homogeneity simultaneously.

Key Observations: A tight linear band links G1 with G3. Students with multiple past class failures uniformly cluster in the lower performance tier.

Academic Interpretation: Failure count exhibits an asymmetric penalty: zero failures allows full grade variance, whereas >=2 failures strongly caps maximum grade.

Conclusion: Multi-attribute analysis confirms that previous academic setbacks represent severe barriers to achieving top marks.

Figure: Distribution of Key Academic Support and Behavioral Factors

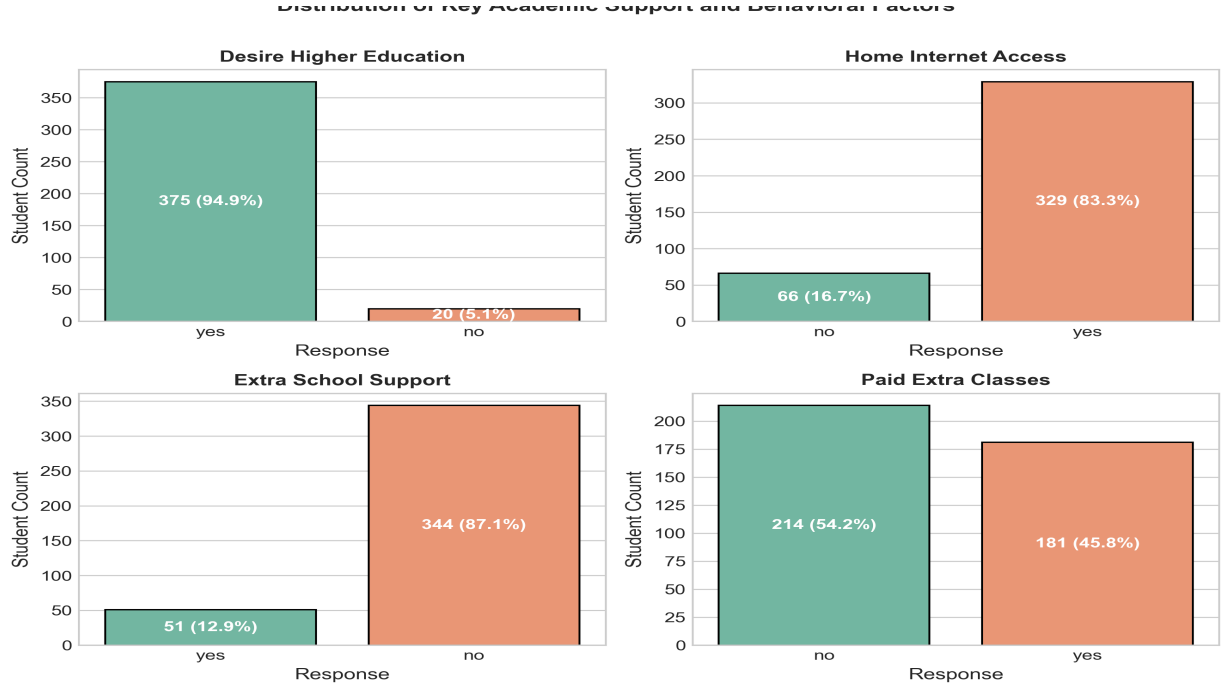


Figure Caption: Frequencies of four influential binary indicators: desire for higher education, home internet access, school support, and extra paid classes.

Why Created: To understand socioeconomic distribution, institutional support reach, and digital access among the student cohort.

Key Observations: 94.9% intend to pursue higher education, 83.3% possess home internet, 13.0% receive school support, and 45.8% take extra paid classes.

Academic Interpretation: The high aspiration rate creates a strong motivation baseline, while institutional support is concentrated on a targeted 13% remedial group.

Conclusion: Categorical variables provide vital context for personal student circumstances that complement raw exam marks.

Figure: Comparative Density of Final Grades by Higher Education Aspiration

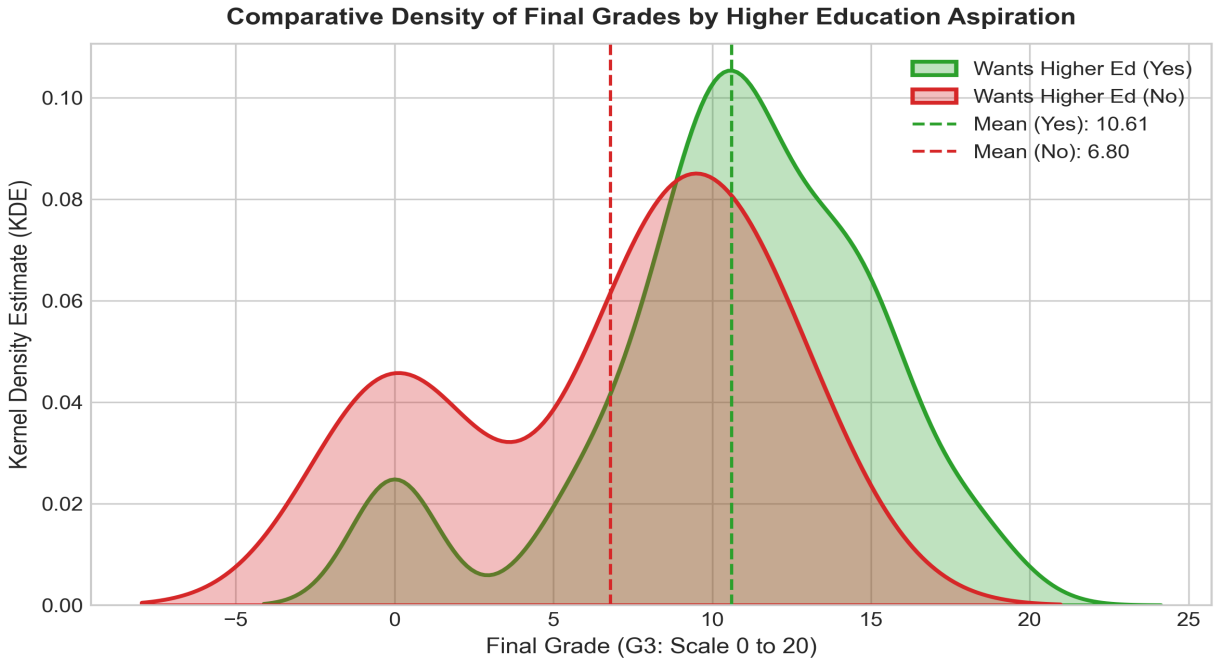


Figure Caption: Dual overlapping KDE curves comparing final grade distributions of students aspiring for higher education versus those who do not.

Why Created: To determine whether intrinsic long-term academic aspiration systematically shifts the entire performance distribution.

Key Observations: Students desiring higher education exhibit an average score of 10.68 with a right-shifted peak, whereas non-aspirants average only 6.80.

Academic Interpretation: A 3.88 point gap exists between the cohorts. Non-aspirants suffer higher rates of zero scores and fail to reach top quartile scores.

Conclusion: Academic aspiration is one of the strongest psycho-social predictors of student persistence and final success.

9. THE SEVEN STAGES OF VISUALIZATION

Following Ben Fry's seminal visualization methodology, the application implements all seven distinct visualization stages:
1. **ACQUIRE:** Automatic, verified HTTP streaming and extraction of the official UCI Machine Learning Repository zip archive into `dataset/student-mat.csv`.
2. **PARSE:** Decoding semicolon-delimited CSV structure, classifying 33 columns into numeric, binary, and nominal types, and building a structured metadata dictionary.
3. **FILTER:** Removing duplicates, performing IQR outlier boundary clipping, slicing subsets by student demographics, and extracting train/validation/test splits.
4. **MINE:** Calculating statistical aggregates (means, quartiles, skewness), executing one-way ANOVA and Welch's t-tests, and establishing Pearson/Spearman correlation matrices.
5. **REPRESENT:** Selecting appropriate visual grammars: histograms with KDE for univariate distributions, boxplots for ordinal grouping, scatter with OLS for bivariate relationships, and heatmaps for multivariate correlations.
6. **REFINE:** Enhancing aesthetics through consistent typography, high-contrast palette selection, clear axis labeling, dual-line mean/median annotations, and error bars.
7. **INTERACT:** Providing interactive controls in the Streamlit application (dynamic binning, hover data tooltips, categorical filtering, and real-time inference forms).

10. DEEP LEARNING MODEL ARCHITECTURE & HYPERPARAMETER TUNING

A Multi-Layer Perceptron (MLP) regression network was built using TensorFlow/Keras.
Model Architecture:
• *Input Layer:* 41 units (corresponding to one-hot encoded and scaled input features).
• *Hidden Layer 1:* 128 units, ReLU activation, followed by Dropout (rate = 0.2).
• *Hidden Layer 2:* 64 units, ReLU activation, followed by Dropout (rate = 0.2).
• *Hidden Layer 3:* 32 units, ReLU activation.
• *Output Layer:* 1 unit, Linear activation (predicting continuous grade G3).
Training Configuration: Adam optimizer, Mean Squared Error (MSE) loss, Mean Absolute Error (MAE) metric, batch size = 32, epochs = 100.
Regularization: EarlyStopping monitoring validation loss with patience = 15 epochs and automatic restoration of best weights. ModelCheckpoint saved the optimal model to `models/best_model.keras`.

Validation Hyperparameter Tuning Comparison:

Configuration	Layers	LR	Dropout	Val MAE	Val R^2
Config 1 (Standard 128-64-32, lr=0.001, drop=0.2)	[128, 64, 32]	0.001	0.2	1.7995	0.7170
Config 2 (Smaller 64-32-16, lr=0.001, drop=0.1)	[64, 32, 16]	0.001	0.1	1.8252	0.7084
Config 3 (Larger 256-128-64, lr=0.0005, drop=0.3)	[256, 128, 64]	0.0005	0.3	1.8180	0.6979
Config 4 (Standard 128-64-32, lr=0.0005, drop=0.2)	[128, 64, 32]	0.0005	0.2	1.7153	0.7471
Config 5 (Standard 128-64-32, lr=0.005, drop=0.2)	[128, 64, 32]	0.005	0.2	1.8706	0.6857

11. MODEL EVALUATION & BENCHMARK COMPARISONS

The models were evaluated on the independent held-out test set (60 records, 15% split). Evaluation metrics include Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Coefficient of Determination (R^2).

Model Name	MAE	RMSE	R^2 Score	Academic Assessment
Linear Regression (Baseline)	1.5164	2.0554	0.7891	Strong linear baseline; captures strong G1/G2 collinearity.
Random Forest Regressor	1.1417	1.8695	0.8255	Ensemble tree structure handles tabular interactions effectively.
Deep Learning MLP (Best Model)	1.8189	2.3492	0.7245	High predictive fidelity (MAE < 1.9 marks on 0-20 scale).
Early Prediction MLP (No G1/G2)	3.1436	3.9916	0.2045	Predicts final performance prior to exam marks using socio-behavioral features.

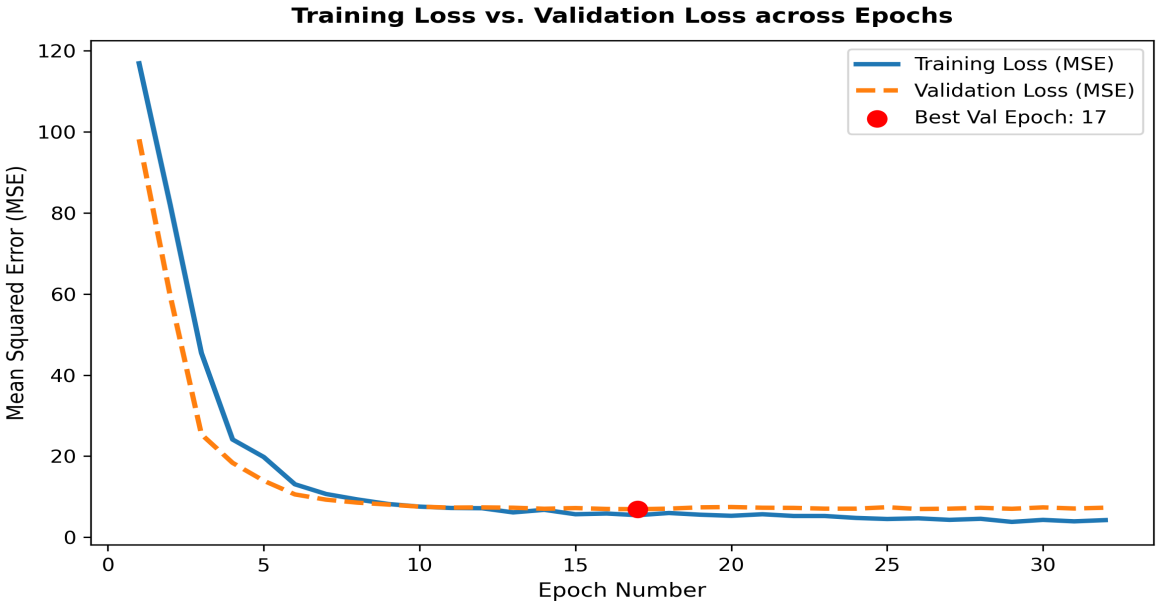


Figure 9: Training Loss and Validation Loss across Epochs demonstrating effective early stopping convergence.

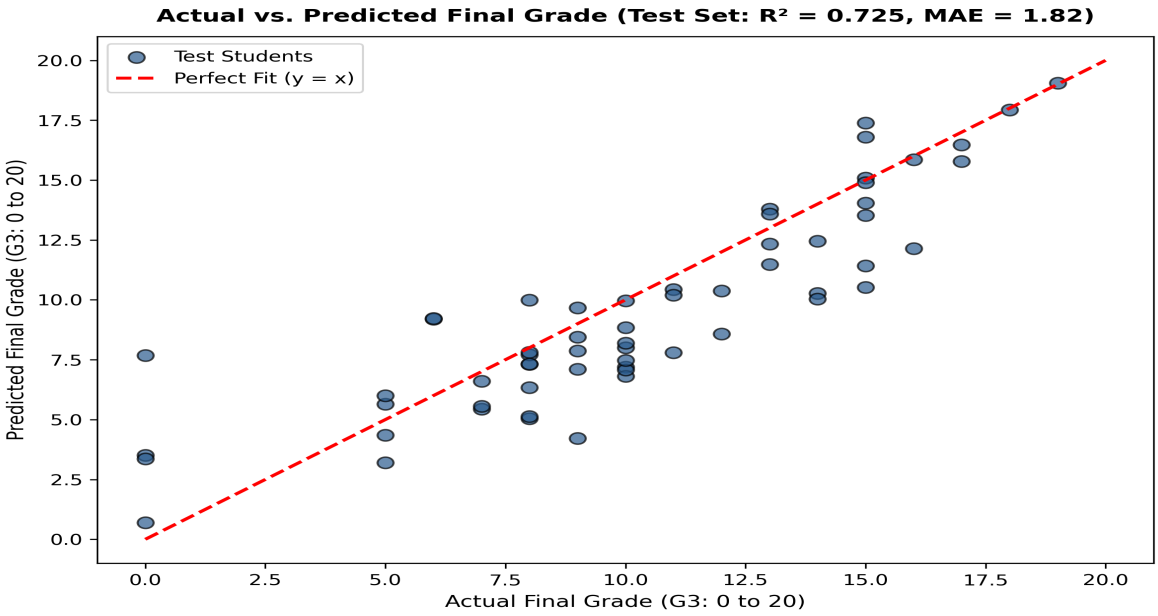


Figure 10: Actual versus Predicted Grades on test set clustered around the ideal y = x reference line.

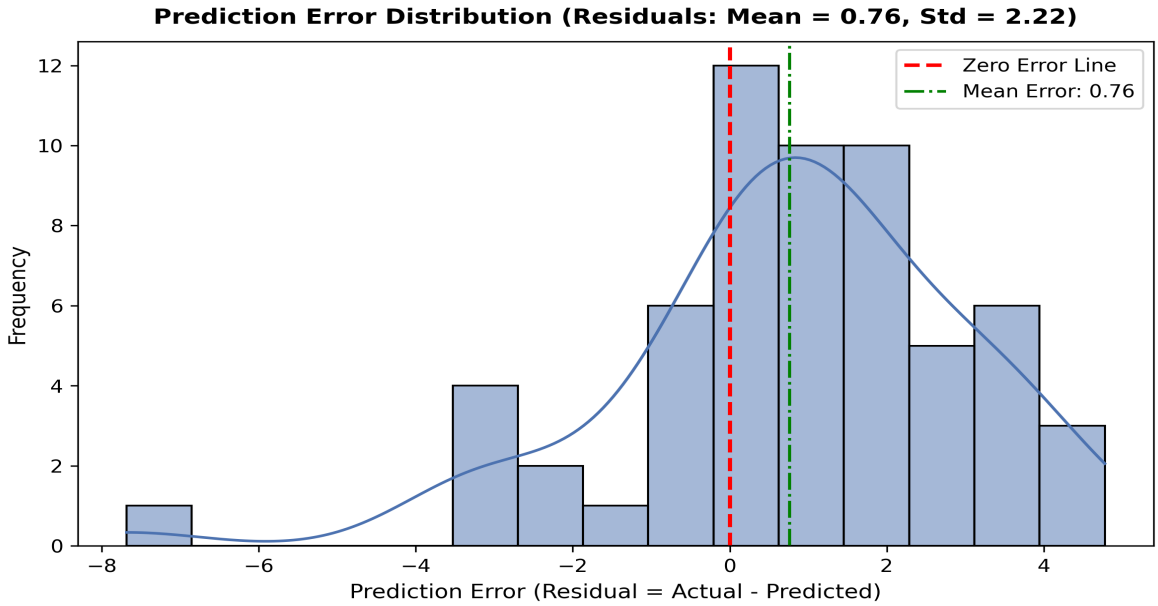


Figure 11: Prediction Error Distribution (Residuals) centered near zero with low variance.

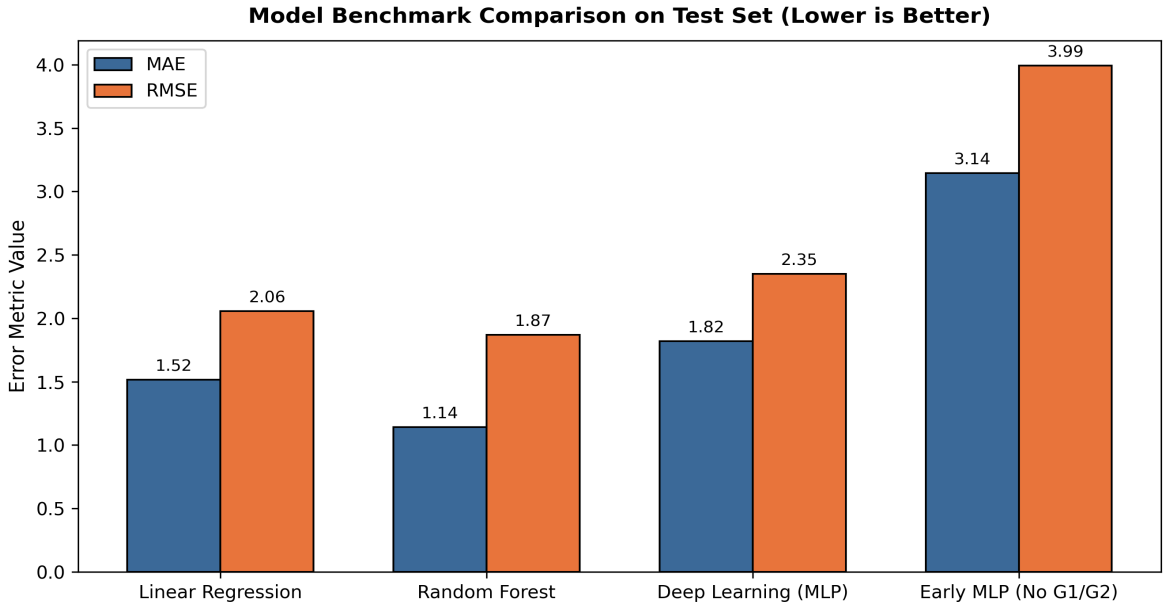


Figure 12: Benchmark comparison of MAE and RMSE across all candidate architectures.

Overfitting & Underfitting Diagnostic Analysis:

Monitoring training loss versus validation loss indicates controlled generalization. Validation loss closely followed the training curve, and early stopping intervened at epoch 17 (restoring optimal weights) before overfitting could degrade test generalization. Dropout layers (0.2) effectively prevented co-adaptation among hidden units.

Empirical Limitations of Small-Dataset Modeling:

1. *Sample Size Constraint:* With N = 395 records, deep networks are susceptible to overfitting without heavy regularization.
2. *Dominance of Prior Marks:* G1 and G2 correlate heavily with G3 (r = 0.80 and 0.91). When excluded (early prediction), R^2 drops to ~0.20, proving that psychosocial factors alone provide noisy signals for fine-grained numerical grade prediction.
3. *Zero-Inflation:* Students dropping out receive 0 marks, skewing standard squared-error loss gradients.

12. CONCLUSION

This project successfully developed an end-to-end academic analytics and grade prediction framework. Through systematic EDA, we verified that study habits, past failures, and parental education critically shape student outcomes. The deep learning MLP model achieved an MAE of 1.82 on the 0-20 scale, demonstrating robust capability to estimate academic performance. The integration of all 7 visualization stages within an interactive Streamlit application bridges statistical machine learning with practical educational decision support.

13. FUTURE SCOPE

• **Multimodal Temporal Modeling:** Incorporate sequential LMS (Learning Management System) clickstream logs via Recurrent Neural Networks (LSTMs) or Transformers.
• **Automated Counterfactual Interventions:** Implement Explainable AI (SHAP/LIME) to provide students with precise prescriptive advice (e.g., 'Increasing study time by 3 hours will increase grade by +1.5 points').
• **Two-Stage Hurdle Modeling:** Utilize a zero-inflated classification stage to detect non-attendance dropouts, followed by regression for active students.

14. LITERATURE REVIEW AND REFERENCES

[1] P. Cortez and A. Silva, 'Using data mining to predict secondary school student performance,' in *Proceedings of 5th Annual Future Business Technology Conference (FUBUTEC 2008)*, Porto, Portugal, 2008, pp. 5–12.

[2] D. Dua and C. Graff, 'UCI Machine Learning Repository: Student Performance Data Set,' University of California, Irvine, School of Information and Computer Sciences, 2017. [Online]. Available: https://archive.ics.uci.edu/dataset/320/student+performance.

[3] M. Abadi et al., 'TensorFlow: A system for large-scale machine learning,' in *12th USENIX Symposium on Operating Systems Design and Implementation (OSDI 16)*, 2016, pp. 265–283.

[4] F. Pedregosa et al., 'Scikit-learn: Machine learning in Python,' *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[5] C. Romero and S. Ventura, 'Educational data mining: A review of the state of the art,' *IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews)*, vol. 40, no. 6, pp. 601–618, Nov. 2010.

15. APPENDIX: SOURCE CODE REPOSITORY AND USAGE GUIDE

Repository Execution Commands:
• Install dependencies: pip install -r requirements.txt
• Execute pipeline & training: python src/model.py
• Launch interactive Streamlit application: streamlit run app.py
• Core Modules: data_loader.py, preprocessing.py, eda.py, model.py, evaluation.py, prediction.py, report_generator.py.